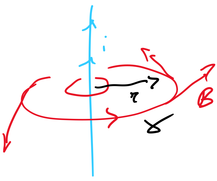
Campi magnetici notevoli

2/5/21 10:58 AM

FILO INDEFINITO



i stazionaria

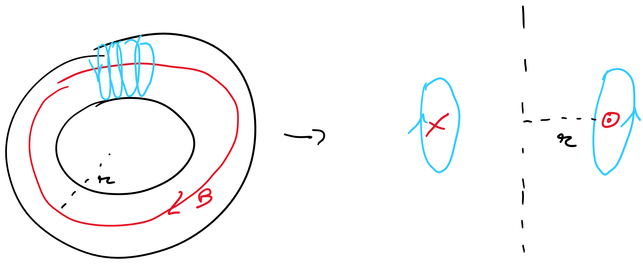
$\nabla \times B = \mu_0 \vec{J}$

$\oint_\gamma \vec{B} \cdot d\vec{l} = \mu_0 I \qquad \vec{B} = B \hat{u}_\varphi$

$B\, 2\pi r = \mu_0 I$

Legge di Biot Savart $\quad \vec{B} = \dfrac{\mu_0 I}{2\pi r} \hat{u}_\varphi$

Solenoide Toroidale



$B_{ext} = 0$ (Non concatena correnti)

$\oint_\gamma \vec{B}\, d\vec{l} = \mu_0 N I \qquad \vec{B} = B\, \vec{u}_\varphi$

$2\pi r B = \mu_0 N I$

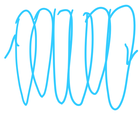
$\vec{B} = \dfrac{\mu_0 N I}{2\pi r} \vec{u}_\varphi$

Solenoide indefinito:

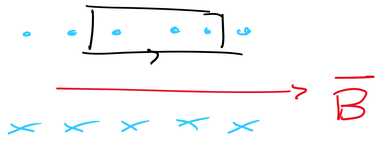
Toro $r \to +\infty$

$B_{int} \parallel$ asse

$B_{ext} = 0$



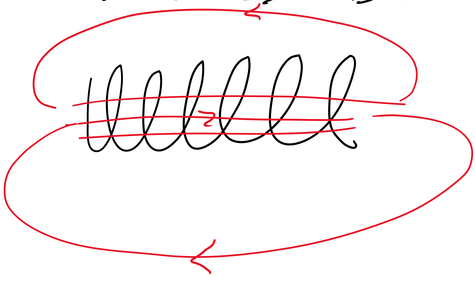
$B = 0$



$\ell B = \mu_0 N I \qquad N/\ell = n$

$\vec{B} = \dfrac{\mu_0 N I}{\ell} \vec{u}_t = \mu_0\, n\, i\, \vec{u}_t$

Solenoide finito

$B_{ext} < B_{int}$

Se $\ell \gg r \Rightarrow B = B_\infty$